Dwijesh Dookraz

SOFTWARE ENGINEER · BACKEND · AI SYSTEMS · APPLIED MACHINE LEARNING

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EDUCATION

University of Southampton *2021 – 2024*

BSc Computer Science · First Class Honours

Dissertation: **Machine Learning-Based Heart Rate Measurement Using rPPG — 82%**

Machine Learning Technologies **80%** Social Computing Techniques **83%** Cloud Application Development **79%**
Software Engineering Group Project **73%**

EXPERIENCE

Nusmark *2024 – Present*

Backend Engineer

- Built bidirectional Google Calendar and Outlook sync (Python, OAuth2, FastAPI) — webhook subscriptions, sync token refresh, and Firestore transactions to eliminate duplicate events in production
- Developed WhatsApp AI assistant (Python, Flask, Cloud API) with NLP task extraction across text, audio, and image — deployed on GCP Cloud Run for automated event creation and onboarding
- Engineered multi-channel notification service (Firestore, Google Calendar, Microsoft Graph) routing reminders and mobile push across cross-provider event lifecycle

PROJECTS

rPPG Heart Rate Estimation · Final-Year Dissertation *2025*

- Implemented DeepPhys dual-stream CNN (PyTorch, OpenCV) for sensor-free heart rate estimation — Pearson r = 0.87, MAE 8.85 BPM on UBFC-rPPG benchmark
- Collected 15-subject dataset (Fitzpatrick II–VI) combined with UBFC-rPPG to address demographic bias — recovering cross-population correlation from r = −0.23 to r = +0.59
- Applied Optuna hyperparameter search with subject-aware k-fold CV, benchmarking against ICA, PCA, and band-pass baselines

Hybrid Recommender System *2026*

- Built hybrid collaborative filtering (user-CF + item-CF + content signals) from scratch in Python/NumPy with streaming SGD and Optuna Bayesian search — MAE 0.587 on 20M MovieLens ratings, no full dataset in RAM

VS Code Portfolio · dwijesh.dev *2024 – 2026*

- Built browser-based VS Code shell (Next.js 15, TypeScript, Monaco Editor) — GitHub API, Sentry monitoring, Vercel Analytics, and Playwright test suite for production deployment
- Designed server-side proxy route to fetch shields.io badges, bypassing CSP restrictions for dynamic badge rendering in the editor preview

Gene Expression Analysis — Osteosarcoma *2025*

- Built scikit-learn pipeline on GSE1000 microarray data — ANOVA selection across 22,283 probes, Shrinkage LDA/LOOCV (accuracy 0.80), Random Forest, and k-means (k=3) confirming 26 ECM genes downregulated via GO enrichment